

Post-2-Year Outcomes After HAART Initiation: Interim Report

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1 Introduction

This study is a secondary analysis of data from the Multicenter AIDS Cohort Study, a prospective cohort examining the progression of HIV-1 infection among homosexual and bisexual men in four U.S. cities.

Among 715 HIV-infected men initiating HAART, we evaluate whether clinical and health-related outcomes improve two years after treatment initiation and whether these changes differ by baseline hard drug use.

We hypothesize that viral load decreases, CD4+ T-cell count increases, and both physical and mental health scores improve after two years of treatment, and that baseline hard drug use attenuates these improvements.

2 Methods

2.1 Data Preparation

We assessed missingness, outliers, and distributional characteristics for all outcomes, exposures, and covariates. Implausible values (e.g., negative CD4 counts, extreme BMI values) were removed. Race and education were recoded to reduce sparse categories. Because viral load was strongly right-skewed, a $\log_{10}$ transformation was applied.

The primary outcome for each measure was defined as:

$$\text{Change} = \text{Year 2 Value} - \text{Baseline Value}.$$

2.2 Exploratory Analysis

Continuous variables were examined using histograms, and correlations were summarized using correlation matrices. Baseline characteristics were compared by hard drug use status using t-tests.

2.3 Regression Analysis

We conducted both frequentist and Bayesian linear regression analyses for four outcomes: change in $\log_{10}$ viral load, CD4+ T-cell count, physical health score, and mental health score.

The primary exposure was baseline hard drug use (yes/no). All models adjusted for baseline age, race, education, BMI, depression score, alcohol use, and smoking status.

Frequentist models were estimated using ordinary least squares. Regression coefficients, 95% confidence intervals, and p-values were reported.

Bayesian models were implemented in Stan with weakly informative priors. Posterior means, 95% credible intervals, and posterior probabilities were summarized.

Model reduction used backward selection based on AIC (frequentist) and WAIC (Bayesian). Consistency between modeling approaches was assessed.

For Bayesian models, convergence diagnostics included trace plots, $\hat{R}$ statistics, effective sample size, and E-BFMI (threshold > 0.3).

3 Preliminary Results

Of 715 men initiating HAART, 506 had both baseline and two-year follow-up data. Missingness for individual variables was low ($< 5\%$).

Histograms of the transformed outcomes are shown below. The distributions were approximately normal, supporting the use of linear regression models.

